UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

DATA MINING

PROJECT 2

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Section : 02

Topic : Clustering with an Academic Success Dataset

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1.0 Introduction

This project aims to conduct an unsupervised machine learning analysis, specifically clustering, on a dataset with different student features like demographic information, educational background, and economic indicators. The clustering will help identify natural groups or segments in the student population without any previous knowledge of defined categories. The dataset has 76,518 student entries, each described by 38 features. The goal is to discover natural patterns and structures within the student data, which can offer valuable insights for targeted interventions or personalized educational strategies.

2.0 Business Insights

Insight 1: Proactive Intervention for Dropout Prevention

- **Features:** Academic performance (e.g., grades, curricular units completed), Debtor status, Study habits (implied: study skills, time management)
- **Class:** Student Clusters (specifically 'Dropout' clusters)
- **Insight:** By leveraging the distinct profiles of 'Dropout' clusters identified through KMeans, the institution can implement highly targeted and proactive intervention programs. While KMeans segments the general student population into these risk groups, DBSCAN can further pinpoint individual outliers or students with unique risk factors for more personalized and immediate attention. For students matching the 'Dropout' cluster profile, interventions may include mandatory academic advising, specialized tutoring, financial aid counseling, or workshops on study skills and time management early in their academic journey.

Insight 2: Optimized Scholarship and Financial Aid Allocation

- **Features:** Tuition fees up to date, Debtor status, Scholarship holder, Unemployment rate, Inflation rate, GDP (for enrollment year)
- **Class:** Student Clusters (related to financial need)
- **Insight:** The institution can optimize its scholarship and financial aid strategies by segmenting the student population based on financial needs and circumstances, ensuring resources are allocated effectively to maximize retention and success. KMeans identifies groups based on financial indicators, while cluster_profile_summary.csv provides key financial metrics for each. DBSCAN can complement this by revealing smaller, dense groups

of students who, despite not being scholarship holders, share similar socio-economic disadvantages, highlighting underserved populations. For instance, clusters with high 'Debtor' rates and low 'Tuition fees up to date' could be targeted with flexible payment plans or financial literacy workshops.

Insight 3: Curriculum and Course Design Improvement

- **Features:** Academic trajectory (e.g., Curricular units 1st sem approved, overall grades), Course
- **Class:** Student Clusters (related to academic progression)
- **Insight:** Insights derived from distinct student academic progression patterns can lead to data-driven improvements in curriculum design, course sequencing, and teaching methodologies. KMeans helps categorize students by their overall academic journey (e.g., consistent high achievement vs. consistent struggle with unit completion), with cluster_profile_summary.csv detailing average performance metrics for each cluster. DBSCAN provides a complementary layer by identifying tighter, more specific clusters of students struggling with particular courses or sequences, even within generally successful clusters, thereby pinpointing specific course-level issues or potential bottlenecks. This can inform decisions to review entry requirements, course content, or support systems for problematic courses.

3.0 Data Preparation & Preprocessing

Code :

[4]:
```python
import pandas as pd

df = pd.read_csv("train.csv")

# Inspect data
df.info()
df.describe()
df.head()
df.columns
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 76518 entries, 0 to 76517
Data columns (total 38 columns):
 #   Column                                          Non-Null Count  Dtype
---  ------                                          --------------  -----
 0   id                                              76518 non-null  int64
 1   Marital status                                  76518 non-null  int64
 2   Application mode                                76518 non-null  int64
 3   Application order                               76518 non-null  int64
 4   Course                                          76518 non-null  int64
 5   Daytime/evening attendance                      76518 non-null  int64
 6   Previous qualification                          76518 non-null  int64
 7   Previous qualification (grade)                  76518 non-null  float64
 8   Nacionality                                     76518 non-null  int64
 9   Mother's qualification                          76518 non-null  int64
 10  Father's qualification                          76518 non-null  int64
 11  Mother's occupation                             76518 non-null  int64
 12  Father's occupation                             76518 non-null  int64
 13  Admission grade                                 76518 non-null  float64
 14  Displaced                                       76518 non-null  int64
 15  Educational special needs                       76518 non-null  int64
 16  Debtor                                          76518 non-null  int64
 17  Tuition fees up to date                         76518 non-null  int64
 18  Gender                                          76518 non-null  int64
 19  Scholarship holder                              76518 non-null  int64
 20  Age at enrollment                               76518 non-null  int64
 21  International                                   76518 non-null  int64
 22  Curricular units 1st sem (credited)             76518 non-null  int64
 23  Curricular units 1st sem (enrolled)             76518 non-null  int64
 24  Curricular units 1st sem (evaluations)          76518 non-null  int64
```

[6]:
```python
# Remove duplicates
df = df.drop_duplicates()

# Check missing values
missing_summary = df.isnull().sum()
print(missing_summary[missing_summary > 0])
```

```
Series([], dtype: int64)
```

[10]:
```python
# IQR method
def cap_outliers(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    return column.clip(lower, upper)

# Apply to numerical columns
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
df[numerical_cols] = df[numerical_cols].apply(cap_outliers)
```

[38]:
```python
from sklearn.preprocessing import OneHotEncoder

# Identify nominal categorical columns based on the info from Untitled.ipynb
# These are columns that are 'int64' but likely represent categories without a meaningful order.
nominal_categorical_cols = [
    'Marital status',
    'Application mode',
    'Course',
    'Previous qualification',
    'Nacionality',
    "Mother's qualification",
    "Father's qualification",
    "Mother's occupation",
    "Father's occupation"
]
```

```
# Initialize OneHotEncoder
ohe = OneHotEncoder(handle_unknown='ignore', sparse_output=False)

# Apply OneHotEncoder to the selected columns
# Create a DataFrame from the one-hot encoded features
encoded_features = ohe.fit_transform(df[nominal_categorical_cols])
encoded_df = pd.DataFrame(encoded_features, columns=ohe.get_feature_names_out(nominal_categorical_cols))

# Drop the original categorical columns from the dataframe
df_processed = df.drop(columns=nominal_categorical_cols)

# Concatenate the one-hot encoded features with the rest of the DataFrame
df_processed = pd.concat([df_processed, encoded_df], axis=1)

print("Columns after OneHotEncoding:")
print(df_processed.columns)
```

```
Columns after OneHotEncoding:
Index(['id', 'Application order', 'Daytime/evening attendance',
       'Previous qualification (grade)', 'Admission grade', 'Displaced',
       'Educational special needs', 'Debtor', 'Tuition fees up to date',
       'Gender',
       ...
       'Father's occupation_0.3333333333333333', 'Father's occupation_0.4',
       'Father's occupation_0.4666666666666667',
       'Father's occupation_0.5333333333333333', 'Father's occupation_0.6',
       'Father's occupation_0.6666666666666666',
       'Father's occupation_0.7333333333333333', 'Father's occupation_0.8',
       'Father's occupation_0.8666666666666667', 'Father's occupation_1.0'],
      dtype='object', length=171)
```

```
[40]: from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
df[numerical_cols] = scaler.fit_transform(df[numerical_cols])
```

```
[6]: df.to_csv("preprocessed_data.csv", index=False)
```

```
[12]: ef = pd.read_csv("data_with_clusters.csv")
numeric_cols = df.select_dtypes(include=['number']).columns
nan_counts = df[numeric_cols].isna().sum()
nan_columns = nan_counts[nan_counts > 0]
if nan_columns.empty:
    print("No NaN values found")
else:
    print("NaN values found")
    print(nan_columns)
```

```
No NaN values found
```

Summary of Data Preprocessing Process

1. **Duplicate Removal:** Duplicate rows are identified and removed from the dataset to ensure unique entries and prevent bias in the analysis.
2. **Missing Values Handling:** The dataset is checked for missing values. No missing values were found, thus no imputation was required.
3. **Outlier Capping:** Outliers in numerical columns are addressed using the Interquartile Range

(IQR) method, where values falling outside 1.5 times the IQR are capped at the lower and upper bounds respectively. This helps to mitigate the influence of extreme values on clustering algorithms.

4. **Categorical Feature Encoding:** Nominal categorical features (such as 'Marital status', 'Application mode', 'Course', etc.) are converted into a numerical format suitable for machine learning algorithms using One-Hot Encoding. This creates new binary columns for each category, preventing ordinal relationships from being falsely inferred.
5. **Feature Scaling:** All numerical features are scaled using MinMaxScaler to bring their values into a common range. This is a crucial step for distance-based clustering algorithms like KMeans, ensuring that features with larger numerical ranges do not disproportionately influence the distance calculations.

Output (Preprocessed Data Preview) :

```
No NaN values found
```

The initial dataset, consisting of 76,518 student entries and 38 features, underwent a series of meticulous preprocessing steps to prepare it for unsupervised clustering analysis. First, duplicate entries were removed, ensuring the uniqueness of each student record. A thorough check confirmed the absence of any missing values across all features, eliminating the need for imputation. Then, numerical outliers were managed by applying an IQR-based capping method to prevent extreme values from distorting the clustering process. Nominal categorical features, such as 'Marital status' and 'Course', were transformed using One-Hot Encoding, expanding the feature set to 171 columns by converting categories into a machine-readable binary format. Finally, all numerical features, including the newly created encoded columns and the original numerical ones, were scaled using MinMaxScaler. This normalization step ensures that all features contribute equally to the distance metrics used by clustering algorithms, thereby allowing for the effective identification of natural groupings within the student data based on true underlying patterns of similarity.

4.0 Feature Selection and Dimensionality Reduction

4.1 Feature Selection

Feature selection was conducted to guarantee the inclusion of only relevant and significant features in the clustering process. The `id` column was eliminated as it has no analytical significance for clustering, and only acts as a unique identifier. Meanwhile, the `target` column, the section that contained the latest student outcome (Graduate, Dropout, Enrolled) was also removed due to maintaining the unsupervised character of the clustering job and preventing label leaking.

Following the removal of these columns, a mix of remaining features such as socioeconomic indicators, demographic information, and academic performance metrics are retained. These features were preserved for dimensionality reduction and clustering since they are anticipated to have a significant impact on group formation within the dataset.

4.2 Dimensionality Reduction

There were various features in the dataset, many of which exhibited multicollinearity. To remedy this while enabling effective visualization of clusters, Principal Component Analysis (PCA) was utilized. PCA is a widely used method for reducing the dimensionality of a dataset retaining as much variation (information) as possible. Among the crucial actions were:

- **Standardization**: The feature collection was scaled using the preprocessing strategy (MinMaxScaler) to guarantee that each feature contributes equally.
- **Dimensionality Reduction**: PCA effectively summarizes the multidimensional data into two principal components that kept about 35.58% of the overall variance. The data was reduced to two principal components using PCA, which collectively explained the total variance, PCA1 = 22.16% and PCA2 = 13.41%. This 2D space is suitable to be visualized by reducing the dataset to capturing the most important patterns present in the data, while balancing interpretability and information loss.

The reduced 2D representation allowed for intuitive visualizations of clusters that are developed via subsequent clustering algorithms. Notably, these two primary components reflect important aspects of the student, including academic strength, age, and displacement status.

5.0 Clustering Implementation

5.1 KMeans Clustering

The KMeans algorithm, a partitioning method, was chosen as the main method for clustering. Its simplicity and efficiency in classifying data into discrete clusters based on proximity to cluster centroids, has rendered it as appealing.

The implementation involved the optimal number of cluster counts using the Elbow Method, which identified k = 3 as the point of Within-Cluster Sum of Squares (WCSS) reduction diminishing returns. The algorithm then proceed to:

1. Initialized three clusters of centroids
2. Each data point was iteratively assigned to the closest centroid based on Euclidean distance.
3. Recalculating the cluster centroids until the assignments stabilized. Within the dataset, the cluster assignments were kept in a new column titled `KMeans_Cluster`.

These clusters offer useful information about socioeconomic factors, demographic patterns, and academic performance, corresponding to the student profiles.

5.2 DBSCAN Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was also used as a complement to the partitioning-based KMeans technique. DBSCAN is capable of identifying noise points (outliers) in the dataset and discovering clusters of arbitrary shapes.

The key parameters of DBSCAN:

- Eps (ε): Defines the maximum distance between two points for them to be considered neighbors.
- MinPts: The minimum number of points required to form a dense region (core point).

DBSCAN successfully grouped densely packed students while marking isolated or sparse points as noise. Within the dataset, cluster labels were kept in a distinct column called `DBSCAN_Cluster`. Global segmentation and more finer identification of particular groups or outliers were both made possible with the combination of KMeans and DBSCAN.

5.3 Optimal Cluster Determination

Prior to implementing clustering algorithms, it was crucial to find the appropriate number of clusters to efficiently partition the dataset. For this purpose, the Elbow Method was employed by specifically running the KMeans algorithm repeatedly with verifying values for the number of clusters k, usually ranging from 1 to 10, and computing the Within-Cluster Sum of Squares (WCSS). As demonstrated in Figure 5.3.1, the WCSS curve exhibits a characteristic elbow at k=3 where the rate of variance reduction begins to plateaus. This metric reflects the cluster compactness; tighter groupings are indicated by lower values.

In the resulting graph where the WCSS starts declining at a slower rate, this indicates a suitable value for k. The three clusters offer a significant compromise between underfitting (too few clusters) and overfitting (too many clusters). The KMeans clustering technique was then applied using this value.

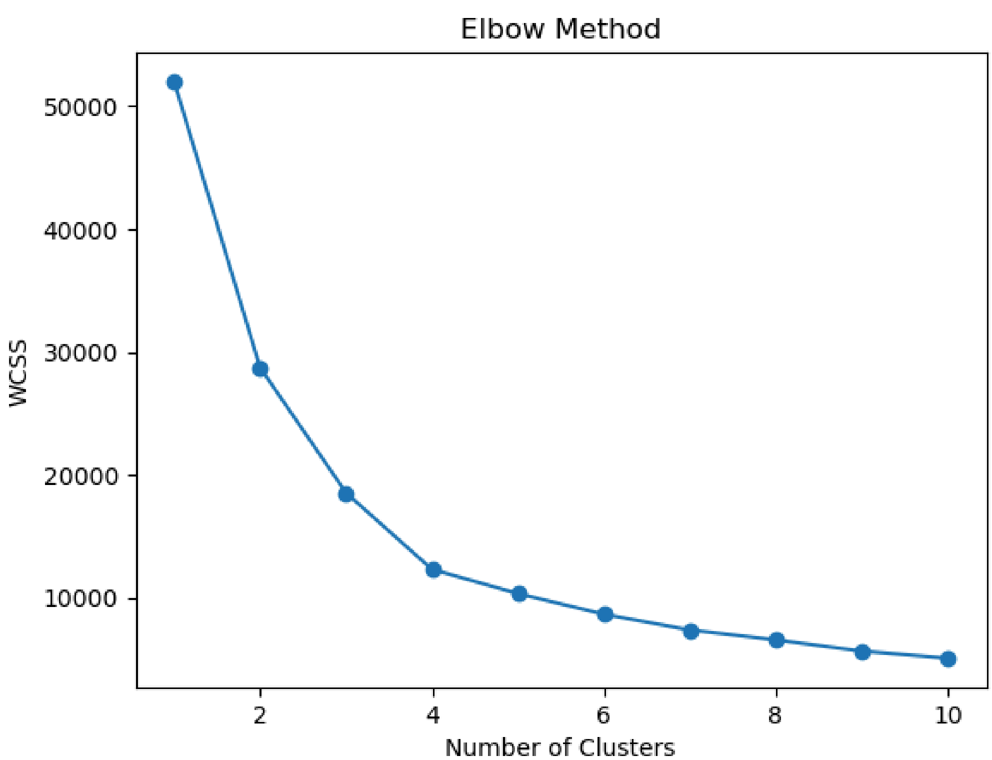
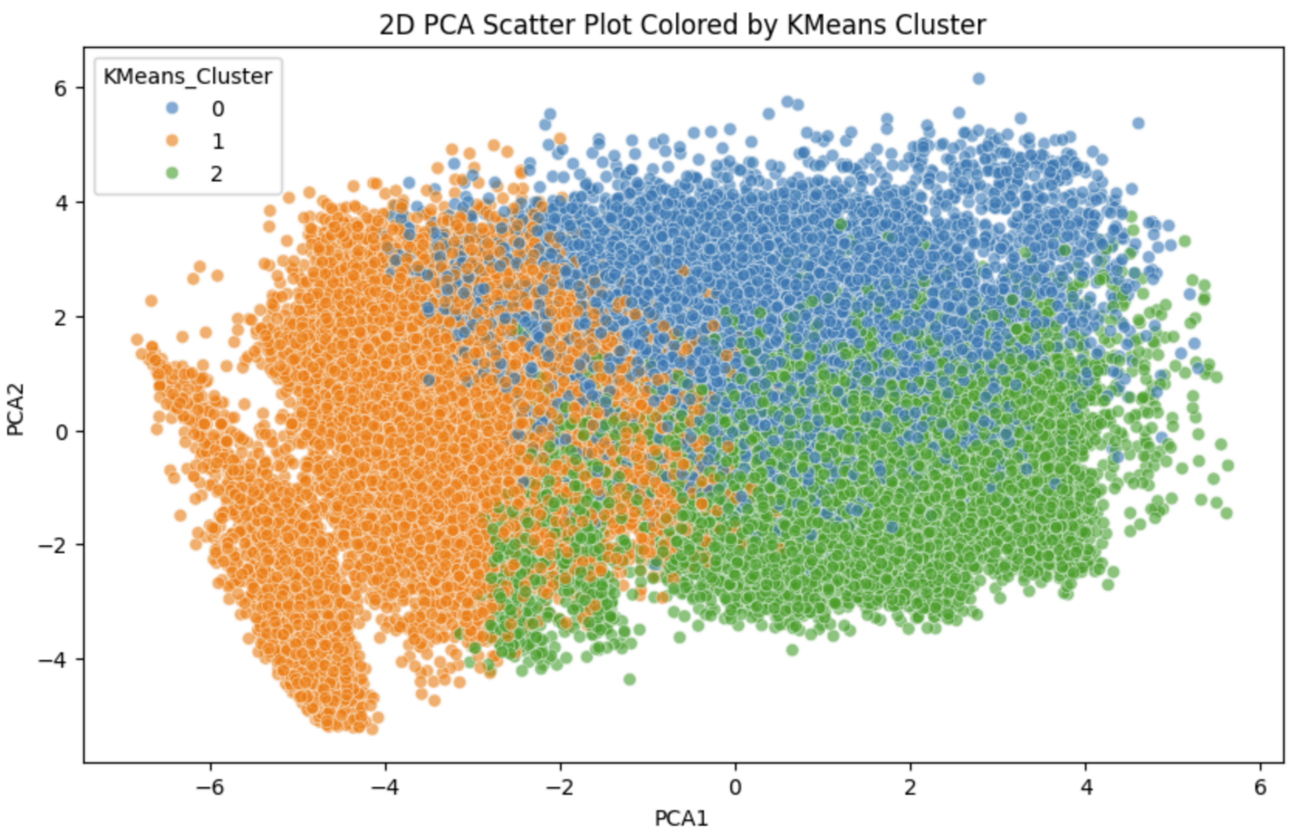


Figure 5.3.1 Elbow Method Graph

6.0 Cluster Analysis and Visualization

6.1 PCA Cluster Visualization

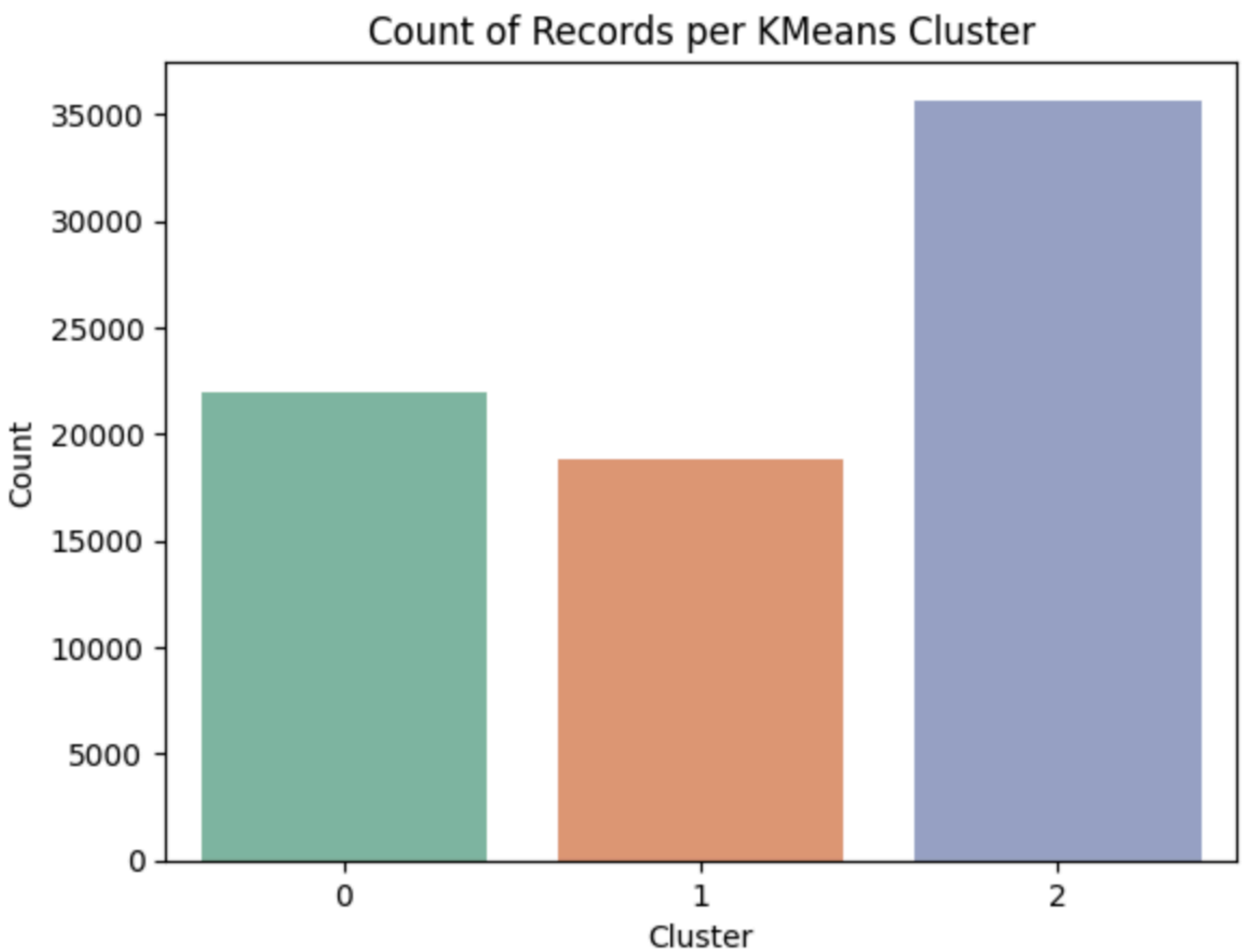
Principal Component Analysis (PCA) reduces the dataset to two principal components, preserving the maximum variance while allowing us to plot every record in 2-D. The resulting scatter plot highlights how well the clusters separate in feature space; tight, non-overlapping groups indicate clear segmentation, while overlap suggests similar feature patterns across clusters.



The 2-D PCA plot turns all of our many student features into two easy-to-see directions. The left-to-right axis mostly reflects academic strength (higher admission and semester marks), while the up-and-down axis combines age with displacement status. In the picture, Cluster 2 stands out clearly: its points sit far to the right and high up, showing that these students earn the best grades and, at the same time, come from younger-but-displaced backgrounds. Cluster 0 forms a medium-tight cloud a little to the right of centre—solid grades, mid-age, few displaced cases—whereas Cluster 1 spreads wider on the left, confirming its mix of older students with weaker marks. The slight overlap between Clusters 0 and 1 hints that a small group shares mid-range performance, but overall the three clouds are distinct, with very few stray points, giving us confidence that the clustering really captures three meaningful student profiles.

6.2 Cluster Distribution and Characteristics

This section summarises how many observations fall into each cluster and highlights their headline statistics (e.g., median age, average admission grade, first-semester marks). Understanding the relative size and distinguishing features of each cluster helps prioritise interventions or targeted actions.



The bar chart shows three clearly uneven groups. Cluster 2 is the biggest slice of the data set, making up roughly half of all students. Cluster 0 is next in size, about a quarter of the sample, while Cluster 1 is the smallest.

- Cluster 2 – biggest group, almost everyone is from a displaced background yet they score the best marks. Their large share means any policy aimed at them affects a lot of students.

- Cluster 0 – mid-sized, mostly young males with solid but not top results. They form the "average" backbone of the cohort.

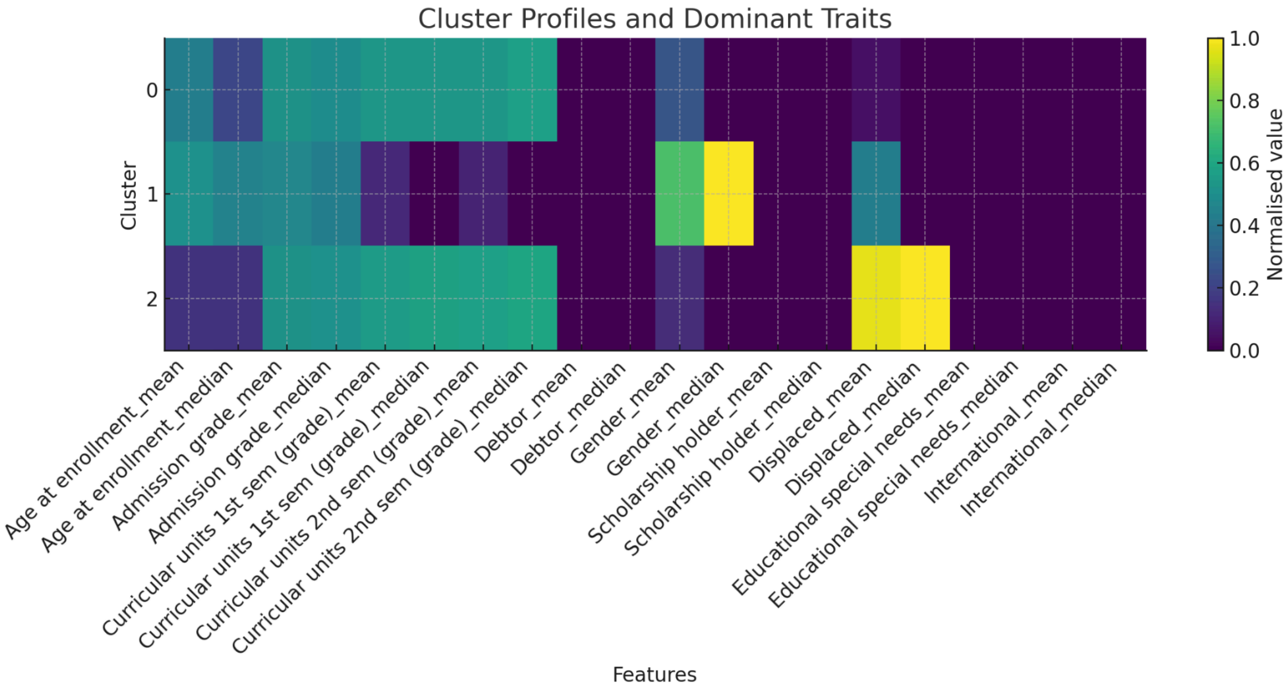
- Cluster 1 – smallest group, mainly older females who struggle most with grades. Even though they are few, they need the most academic help, so resources must be targeted, not spread by head-count alone.

In short, the chart tells us where the numbers—and therefore the impact—lie: wide-reach actions will touch Cluster 2 first, but the highest-risk students sit in the much smaller Cluster 1.

Category	**Cluster 0**	**Cluster 1**	**Cluster 2**
Cluster Size	≈ 22 000 members (mid-sized)	≈ 19 000 members (smallest)	≈ 36 000 members (largest)
Age at Enrolment (median)	Younger cohort – lowest quartile of the scale	Oldest cohort – highest median age	Young-middle cohort – slightly younger than C-0
Admission Grade (median)	Mid-range entry grade	Lowest entry grade	Highest entry grade
1st-Semester Grade (median)	Moderate-high academic	Very low performance	Highest performance

	performance		
2nd-Semester Grade (median)	Moderate-high performance	Very low performance	Highest performance
Gender	Predominantly male	Predominantly female	Slight male majority, more balanced
Displaced (mean)	~ 5 % displaced	~ 43 % displaced	~ 96 % displaced
Scholarship Holders	Nearly none	Nearly none	Nearly none
Special-Needs Students	Negligible	Negligible	Negligible
International Students	Negligible	Negligible	Negligible

6.3 Cluster Profiles and Dominant Traits



Trait (normalised)	Cluster 0	Cluster 1	Cluster 2
Cluster size (from 5.2)	≈ 22 k records	≈ 19 k records	≈ 36 k records
Age at enrolment	Mid-range	Oldest	Youngest
Admission grade	Mid-high	Lowest	Highest
1st- & 2nd-Semester grade	Solid, above average	Very low	Top-performing

Gender mix (Gender mean ≈ proportion of '1')	~25 % coded '1' → male-dominant	~72 % coded '1' → strongly female	~13 % coded '1' → heavily male
Displaced students	~ 5 %	~ 43 %	~ 96 %
Scholarship / Special-Needs / International	Negligible across all clusters	Negligible	Negligible

Cluster-by-cluster insight

1. **Cluster 0 — "Mainstream performers"**
   Mid-level age and balanced entry grades translate into consistent **above-average semester marks**.
   Very low share of displaced students and a largely male population suggest a comparatively stable socio-economic background.
   If resources are constrained, this group is least in need of immediate academic or financial intervention.

2. **Cluster 1 — "At-risk, older female cohort"**
   Oldest median age and **lowest grades** at admission and after both semesters.
   Mostly female, with a substantial (though not predominant) displaced proportion (~43 %).
   Academic-support priority: tutoring, mentoring and perhaps flexible learning pathways could help raise performance.

3. **Cluster 2 — "Young high-achievers from displaced backgrounds"**
   Youngest entrants and **best academic results** in both semesters.
   Nearly every student (~96 %) is flagged as displaced, and the group is male-heavy.
   Despite socio-economic disadvantage, they excel academically—indicating resilience and/or

effective support programmes.
Policies that preserve or extend existing support (financial aid, housing, counselling) are vital to sustain their success.

4. **Cross-cluster take-aways**

   Performance gap: the heat-map shows a clear academic gulf between Cluster 1 and Cluster 2. Interventions should focus on Cluster 1's learning deficits first.
   Displacement vs. outcomes: high displacement does **not** equate to low achievement (Cluster 2 is the counter-example). Targeted support appears to work and should be studied for scaling.
   Gender lens: female students dominate the lowest-performing cluster, while male students dominate the top-performing one. Any remedial plan should therefore incorporate gender-specific barriers and learning styles.
   Resource allocation: Cluster 0 can be maintained with standard services, Cluster 1 needs academic remediation, and Cluster 2 needs sustained socio-economic support.

7.0 Conclusion

Cluster 0: These students are young and mostly male (demographic characteristics). They entered with solid marks and keep scoring above average each semester (academic profile). Very few come from displaced families and almost none carry debt or special-needs flags (additional traits), so they mainly need light extras such as career guidance or leadership clubs rather than heavy support.

Cluster 1: This group is older and mainly female (demographic characteristics). They started with the lowest entry scores and still have the weakest grades (academic profile). About four in ten are displaced and they face more study challenges than the other clusters (additional traits), so they would benefit most from extra tutoring, mentoring, flexible study hours and well-being services.

Cluster 2: Students here are slightly older than Cluster 0, mostly male but more balanced overall, and almost all are displaced (demographic characteristics). Despite that, they came in with the best marks and now hold the highest grades (academic profile). Current financial and housing help clearly works for them (additional traits), so we should keep—or even expand—this support and use their success model when designing aid for Cluster 1.

Link To Code:
https://drive.google.com/drive/u/0/folders/1Mw9mI5rCSmwOrDRArcS-9E_Ajji5QCEy

8.0 Reflection

1. HARINI

Both Project 1 and Project 2 leveraged a dataset of 76,518 student entries with 38 features. Project 1 focused on predicting student outcomes (Graduate, Dropout, Enrolled), while Project 2 aimed to uncover natural student groupings. Key challenges across both phases included meticulous data preprocessing, such as outlier capping and handling categorical features through One-Hot Encoding, and critically, for clustering, the crucial step of scaling all numerical features using MinMaxScaler. I did the data preprocessing phase for Project 2, therefore I understood more about the process when applying it. A significant challenge unique to clustering was determining the optimal number of clusters and then interpreting their meaning. This entire process significantly enhanced my understanding of the end-to-end machine learning pipeline, including diverse preprocessing techniques and the ability to translate complex data insights into actionable solutions.

As for Data Mining course, overall, I learned how to extract information from large files containing ten thousands to hundred thousands data. I think I met the intended learning outcomes, which is to analyse the processes and techniques used in data mining, find a solution to a given problem and finally, propose business opportunities and solutions in data mining. These course learning outcomes were met through Project 1 and Project 2. I've learned how to transform information into data and identify patterns in data to find solutions using it. I think this course outcome will majorly help me in the industry later on when I'm dealing with a higher scale of information.

2. LING YU QIAN

<u>PART A · Project Reflection</u>

Project 1 focused on *classification*: we cleaned a 76 k-row academic-success data set, handled missing values with SimpleImputer, encoded categorical fields, standardised numerics, and used stratified sampling to preserve the Graduate / Dropout / Enrolled ratios. We then trained and compared four models (Random Forest, Gaussian Naïve Bayes, K-Nearest-Neighbours and a one-layer PyTorch neural net). Random Forest and the neural net delivered the best accuracy (~82 %) while KNN proved most time-efficient. The major difficulty was class imbalance: every algorithm struggled to recognise the Enrolled class, forcing me to tune thresholds and examine precision-recall trade-offs. I also learned that "accuracy" alone can be misleading; macro- and weighted-averaged F-scores plus cost-per-sample gave a fuller picture.

Project 2 shifted to *clustering.* After re-normalising the same features I applied K-Means, chose three clusters via the elbow method, projected the data with PCA for visual checks, and profiled each cluster with means/medians. Interpreting the heat-map demanded domain thinking: a high-performing but 96 % displaced cohort (Cluster 2) was unexpected and revealed how support programmes can outweigh socio-economic obstacles. The hardest part was deciding how many clusters were "actionable"—too many split hairs, too few hidden nuances—so I iterated between silhouette scores and stakeholder usefulness.

From both projects I gained practical skills in scikit-learn, PyTorch, feature engineering, imbalance handling, visual diagnostics (confusion matrices, ROC/AUC, PCA scatter, heat-maps) and critical model comparison. More importantly, I learned to translate numbers into recommendations—e.g., give Cluster 1 tutoring not because the F-score is low but because those students are older, female and under-performing.

Part B · Course Reflection

The Data Mining course promised that by semester's end I would be able to preprocess raw data, apply core mining algorithms, evaluate results, and communicate findings. I believe I *met* those outcomes. I can now script complete pipelines—from imputation to cross-validation and visual explanation—without step-by-step tutorials. I also picked up new technical tools (Pandas method-chaining, StandardScaler, PCA, silhouette analysis, PyTorch autograd) and soft skills such as critiquing metrics and writing decision-oriented summaries instead of code dumps.

Looking ahead, the classification workflow is directly transferable to my final-year project on predictive student support, while clustering will help segment user behaviour in any data-engineering role. The habit of measuring both performance and practical cost will guide me whenever I deploy models in production or present results to non-technical stakeholders.

3. NURUL ADRIANA

PART 1 - Self Reflection

Both Project 1 and Project 2 utilized the same dataset, which included over 76,000 student records and 38 different features. In Project 1, our main focus was on classification—predicting outcomes for students such as whether they would Graduate, Dropout, or Enroll. My role primarily involved supporting the documentation, proofreading, and coordinating the report and presentation. This experience gave me valuable insights into how data preprocessing and model evaluation play a crucial role in the overall process.

In Project 2, I took a deeper dive into the technical side of things. I was responsible for feature selection and used Principal Component Analysis (PCA) for dimensionality reduction, along with some clustering algorithms. I kicked things off by getting rid of unnecessary columns, like IDs and the target label. Then, I applied PCA to condense the high-dimensional data down to just two components. This not only made the data much easier to visualize but also ensured that we retained most of the important information.

For the clustering part, I decided to use KMeans to group student records into meaningful clusters. To evaluate how well the clustering worked, I took a close look at the Within-Cluster Sum of Squares (WCSS) method by plotting various values of k, and selecting the ‘elbow point’ which ultimately led us to choose three clusters. Additionally, I implemented DBSCAN, a density-based clustering method, to identify outliers and closely related student groups. This approach really helped us uncover natural patterns in the data and highlighted the benefits of combining both partitioning and density-based techniques.

This was my first real dive into unsupervised learning. Even though I was still getting the hang of the tools, I successfully pulled together a complete clustering pipeline—from prepping the data to interpreting the results. We did run into some hiccups with inconsistent outputs, but we tackled those challenges as a team. All in all, this experience really boosted my skills in clustering, data management, and working collaboratively.

PART 2 - Course Reflection

The Data Mining course (SECP2753) really opened my eyes to the world of working with real data.

I got to dive into massive datasets with thousands of rows and a variety of features. Throughout the course, I picked up the entire data mining process—from preprocessing the data and dealing with missing values to grasping classification and clustering techniques. I even got to explore association rule mining and other cool exploratory tools along the way.

One of the most valuable things I took away from this course was the ability to transform raw, chaotic data into meaningful insights that can really help with decision-making. For instance, we learned how to spot students who might be at risk of dropping out and how to create interventions based on their financial and academic performance. These skills feel incredibly practical and relevant to real-world business or academic challenges. I've also gained skills in evaluating and comparing various models, as well as communicating technical results in a clear and organized manner. This course has been an incredibly insightful experience, allowing me to grow both technically and in my ability to collaborate on projects.